

Questions:

WHERE:

1. The student counselor wanted to display the registration number, student name and date of birth for all the students.
2. The controller of examinations wanted to list all the female students
3. List the Students who registered for the "C001" course.
4. Display all faculty details joined before "November 2014"
5. Display all the courses not allotted to hall 'H001'

LIKE:

6. List the students whose name ends with the substring "sh"
7. Display all students whose name contains the substring "sh"
8. Find all the students who are located in cities having "Sal" as substring
9. Display the students whose names do not contain six letters.
10. Find all the students whose names contains "am".

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mysql> USE CollegeDB;
Database changed
mysql>
mysql> -- STUDENT TABLE
mysql> CREATE TABLE Student
-> (
->     RegNo INT PRIMARY KEY,
->     Name VARCHAR(30),
->     Gender CHAR(1),
->     DOB DATE,
->     City VARCHAR(30)
-> );
Query OK, 0 rows affected (0.08 sec)

mysql>
mysql> -- COURSE TABLE
mysql> CREATE TABLE Course
-> (
->     CourseID VARCHAR(10) PRIMARY KEY,
->     CourseName VARCHAR(50)
-> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> -- REGISTRATION TABLE
mysql> CREATE TABLE Registration
-> (
->     RegNo INT,
->     CourseID VARCHAR(10),
->     PRIMARY KEY(RegNo, CourseID),
->     FOREIGN KEY(RegNo) REFERENCES Student(RegNo),
->     FOREIGN KEY(CourseID) REFERENCES Course(CourseID)
-> );
Query OK, 0 rows affected (0.04 sec)
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mysql>
mysql> -- FACULTY TABLE
mysql> CREATE TABLE Faculty
--> (
-->     FacID INT PRIMARY KEY,
-->     FacName VARCHAR(30),
-->     DOJ DATE,
-->     Department VARCHAR(30)
--> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> -- ALLOTMENT TABLE
mysql> CREATE TABLE Allotment
--> (
-->     CourseID VARCHAR(10),
-->     HallNo VARCHAR(10),
-->     FOREIGN KEY(CourseID) REFERENCES Course(CourseID)
--> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> -- INSERT INTO STUDENT
mysql> INSERT INTO Student VALUES
--> (101, 'Ramesh', 'M', '2004-05-10', 'Salem'),
--> (102, 'Suresh', 'M', '2003-09-12', 'Chennai'),
--> (103, 'Kamesh', 'M', '2004-02-15', 'Salem'),
--> (104, 'Priya', 'F', '2005-01-20', 'Coimbatore'),
--> (105, 'Shyam', 'M', '2003-11-08', 'Salem'),
--> (106, 'Ramya', 'F', '2004-06-18', 'Madurai'),
--> (107, 'Akash', 'M', '2005-07-22', 'Salem'),
--> (108, 'Amala', 'F', '2004-03-14', 'Salem');
Query OK, 8 rows affected (0.01 sec)
Records: 8 Duplicates: 0 Warnings: 0

mysql>
mysql> -- INSERT INTO COURSE
mysql> INSERT INTO Course VALUES

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mysql> -- INSERT INTO COURSE
mysql> INSERT INTO Course VALUES
--> ('C001', 'DBMS'),
--> ('C002', 'Java'),
--> ('C003', 'Python'),
--> ('C004', 'Operating System');
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql>
mysql> -- INSERT INTO REGISTRATION
mysql> INSERT INTO Registration VALUES
--> (101, 'C001'),
--> (102, 'C002'),
--> (103, 'C001'),
--> (104, 'C003'),
--> (105, 'C001'),
--> (106, 'C004'),
--> (107, 'C002'),
--> (108, 'C003');
Query OK, 8 rows affected (0.01 sec)
Records: 8 Duplicates: 0 Warnings: 0

mysql>
mysql> -- INSERT INTO FACULTY
mysql> INSERT INTO Faculty VALUES
--> (1, 'Arun', '2013-05-20', 'CSE'),
--> (2, 'Bala', '2015-02-14', 'ECE'),
--> (3, 'Charan', '2012-10-11', 'IT'),
--> (4, 'Divya', '2014-09-25', 'EEE');
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql>
mysql> -- INSERT INTO ALLOTMENT
mysql> INSERT INTO Allotment VALUES
--> ('C001', 'H001'),
--> ('C002', 'H002'),
--> ('C003', 'H003'),
--> ('C004', 'H002');
Query OK, 4 rows affected (0.01 sec)

```

```

mysql>
mysql> -- QUESTIONS
mysql>
mysql> -- 1. Display the registration number, student name and date of birth for all students.
mysql> SELECT RegNo, Name, DOB
mysql> FROM Student;
+-----+
| RegNo | Name   | DOB       |
+-----+
| 101   | Ramesh | 2004-05-10 |
| 102   | Suresh | 2003-09-12 |
| 103   | Kamesh | 2004-02-15 |
| 104   | Priya  | 2005-01-20 |
| 105   | Shyam  | 2003-11-08 |
| 106   | Ramya  | 2004-06-18 |
| 107   | Aakash | 2005-07-22 |
| 108   | Amala  | 2004-03-14 |
+-----+
8 rows in set (0.01 sec)

mysql>
mysql> -- 2. List all the female students.
mysql> SELECT *
mysql> FROM Student
mysql> WHERE Gender='F';
+-----+
| RegNo | Name   | Gender | DOB       | City       |
+-----+
| 104   | Priya  | F       | 2005-01-20 | Coimbatore |
| 106   | Ramya  | F       | 2004-06-18 | Madurai    |
| 108   | Amala  | F       | 2004-03-14 | Salem     |
+-----+
3 rows in set (0.00 sec)

mysql>
mysql> -- 3. List the students who registered for the course "C001".
mysql> SELECT S.RegNo, S.Name
mysql> FROM Student S
mysql> JOIN Registration R
mysql> ON S.RegNo=R.RegNo

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mysql>
mysql> JOIN Registration R
mysql> ON S.RegNo=R.RegNo
mysql> WHERE R.CourseID='C001';
+-----+
| RegNo | Name   |
+-----+
| 101   | Ramesh |
| 103   | Kamesh |
| 105   | Shyam  |
+-----+
3 rows in set (0.00 sec)

mysql>
mysql> -- 4. Display all faculty details joined before November 2014.
mysql> SELECT *
mysql> FROM Faculty
mysql> WHERE DOJ<'2014-11-01';
+-----+
| FacID | FacName | DOJ       | Department |
+-----+
| 1     | Arun    | 2013-05-20 | CSE         |
| 3     | Charan  | 2012-10-11 | IT          |
| 4     | Divya   | 2014-09-25 | EEE         |
+-----+
3 rows in set (0.00 sec)

mysql>
mysql> -- 5. Display all the courses not allotted to hall 'H001'.
mysql> SELECT *
mysql> FROM Allotment
mysql> WHERE HallNo<>'H001';
+-----+
| CourseID | HallNo |
+-----+
| C002     | H002   |
| C003     | H003   |
| C004     | H002   |
+-----+
3 rows in set (0.00 sec)

```

```
mysql> -- 6. List the students whose name ends with "sh".
mysql> SELECT *
-> FROM Student
-> WHERE Name LIKE '%sh';
+-----+-----+-----+-----+-----+
| RegNo | Name   | Gender | DOB       | City   |
+-----+-----+-----+-----+-----+
| 101   | Ramesh | M       | 2004-05-10 | Salem |
| 102   | Suresh | M       | 2003-09-12 | Chennai |
| 103   | Kamesh | M       | 2004-02-15 | Salem |
| 107   | Aakash | M       | 2005-07-22 | Salem |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql>
mysql> -- 7. Display all students whose name contains "sh".
mysql> SELECT *
-> FROM Student
-> WHERE Name LIKE '%sh%';
+-----+-----+-----+-----+-----+
| RegNo | Name   | Gender | DOB       | City   |
+-----+-----+-----+-----+-----+
| 101   | Ramesh | M       | 2004-05-10 | Salem |
| 102   | Suresh | M       | 2003-09-12 | Chennai |
| 103   | Kamesh | M       | 2004-02-15 | Salem |
| 105   | Shyam  | M       | 2003-11-08 | Salem |
| 107   | Aakash | M       | 2005-07-22 | Salem |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql>
mysql> -- 8. Find all the students who are located in cities having "Sal" as substring.
mysql> SELECT *
-> FROM Student
-> WHERE City LIKE '%Sal%';
+-----+-----+-----+-----+-----+
| RegNo | Name   | Gender | DOB       | City   |
+-----+-----+-----+-----+-----+
| 101   | Ramesh | M       | 2004-05-10 | Salem |
| 103   | Kamesh | M       | 2004-02-15 | Salem |
| 105   | Shyam  | M       | 2003-11-08 | Salem |
+-----+-----+-----+-----+-----+
```

```
+-----+-----+-----+-----+-----+
| RegNo | Name   | Gender | DOB       | City   |
+-----+-----+-----+-----+-----+
| 101   | Ramesh | M       | 2004-05-10 | Salem |
| 103   | Kamesh | M       | 2004-02-15 | Salem |
| 105   | Shyam  | M       | 2003-11-08 | Salem |
| 107   | Aakash | M       | 2005-07-22 | Salem |
| 108   | Amala  | F       | 2004-03-14 | Salem |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql>
mysql> -- 9. Display the students whose names do not contain six letters.
mysql> SELECT *
-> FROM Student
-> WHERE Name NOT LIKE '_____';
+-----+-----+-----+-----+-----+
| RegNo | Name   | Gender | DOB       | City   |
+-----+-----+-----+-----+-----+
| 104   | Priya  | F       | 2005-01-20 | Coimbatore |
| 105   | Shyam  | M       | 2003-11-08 | Salem |
| 106   | Ramya  | F       | 2004-06-18 | Madurai |
| 108   | Amala  | F       | 2004-03-14 | Salem |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql>
mysql> -- 10. Find all the students whose names contain "am".
mysql> SELECT *
-> FROM Student
-> WHERE Name LIKE '%am%';
+-----+-----+-----+-----+-----+
| RegNo | Name   | Gender | DOB       | City   |
+-----+-----+-----+-----+-----+
| 101   | Ramesh | M       | 2004-05-10 | Salem |
| 103   | Kamesh | M       | 2004-02-15 | Salem |
| 105   | Shyam  | M       | 2003-11-08 | Salem |
| 106   | Ramya  | F       | 2004-06-18 | Madurai |
| 108   | Amala  | F       | 2004-03-14 | Salem |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```